COMP1046 Mathematics for Computer Scientists (MCS)
2025-26
Coursework 2

1 Summary.

This coursework is about determinants and inversing matrices. Answer two mathematical questions requiring proof. Write code using NumPy in Python to compute the matrix inverse by I Laplace Theorem.

2 Instructions.

- This coursework has a total of 100 marks and represents 10% of assessment for the COMP1046 module.
- Answer all the questions below.
- Show all your working and use the same notation that is used in the lecture notes. The examples and exercises in the lectures and tutorials are a model for how you should present your answers.
- This must be your own individual work so do not work with other students on this coursework. Do not use AI tools. Ensure you know the University policy on academic misconduct and use of AI:
  - *https://www.nottingham.ac.uk/qualitymanual/assessment-awards-and-deg-classification/pol-academic-misconduct.aspx*
  - *https://moodle.nottingham.ac.uk/mod/folder/view.php?id=8103375*
- You must submit your answers to Q1 & Q2 as a single PDF document `MCS_CW2_`<*student-id*>`.pdf` and for Q3 as a single `.ipynb` file `MCS_CW2_`<*student-id*>`.ipynb` on Moodle. You can use Word or Latex to type up your answer and save as PDF. Alternatively, you can handwrite your answer on A4 paper and scan to PDF. Do not use a mobile phone to take a photo of your coursework and submit the photo, since the quality is often poor.

- Your Python code should run stand-alone on Python3 Jupyter Notebook. You should use the NumPy package, but **no sub-module such as linalg or other packages or external libraries should be used**. If your code does not run or has errors, you could get zero marks for the coding part.
- If your submitted coursework is illegible, you will receive a zero mark for the parts that cannot be read clearly. You will not be offered a second chance to resubmit, so ensure the good quality of your submission.
- Submit by the given deadline **16:00 on 3rd December 2025**. You will lose 5% marks for each day your submission is late.

3 Questions.

For these questions, only use material you have been taught in Lectures 2 to 4 of the linear algebra part of the MCS module. In particular, use the I Laplace Theorem to compute determinant and adjugate matrix, then use these to compute the inverse matrix. For Questions 1 & 2, you may use proof by mathematical induction.

Q1. Prove that for any non-singular square matrix $\mathbf{A}$, $\det \mathbf{A} = \det\left(\mathbf{A}^T\right)$.

Q2. Prove that for any non-singular n order square matrix $\mathbf{A}$, and any scalar $k \in \mathbb{R}$, $\det(k\mathbf{A}) = k^n \det(\mathbf{A})$.

Q3. Write a Python computer program to compute the inverse of a matrix using the I Laplace Theorem described in Lecture 3. Your code should handle the following criteria:

(i) You must write three functions to impement the computation of adjugate matrix, determinant, and inverse of a matrix `A` passed as a NumPy `ndarray` parameter:

- `adjugate(A)` returns `ndarray`;
- `determinant(A)` returns `float`;
- `inverse(A)` returns `ndarray`.

You may use other auxilliary functions but you will be assessed based on these three functions.

(ii) Your code should handle all exception cases when the inverse of `A` cannot be computed, by returning `None` instead of a matrix.

(iii) A *sparse matrix* is one for which over half of its elements are zero. Your code should be written to run computationally efficiently for such matrices. For example, when executing this code, the computation of the inverse of `M2` should be much faster than for `M1`:

```
M1 = np.array([[4,3,2,5,4,5,1],
               [2,1,2,1,2,3,2],
               [4,5,2,5,4,5,1],
               [8,1,2,1,2,3,3],
               [4,3,2,6,4,5,1],
               [2,1,2,1,1,3,2],
               [4,3,2,1,3,4,5]])
print(inverse(M1))
M2 = np.array([[4,0,0,0,0,0,0],
               [0,1,0,1,0,0,0],
               [0,0,2,0,0,0,0],
               [1,0,0,1,0,0,3],
               [0,0,0,0,4,1,0],
               [0,0,0,0,0,3,0],
               [0,0,0,2,0,0,5]])
print(inverse(M2))
```

Please note:

- Your code will be tested in a test script for computing the determinant and inverse of various test matrices. Ensure that your three functions are correctly named to avoid losing marks.
- Your code will be checked to ensure that you have used I Laplace Theorem and the adjugate matrix as described in Lecture 3 to implement your solution.
- Your code will be timed to check for relative difference in performance when processing sparse and non-sparse matrices. The `perf_counter` function in package time is used to do this. You may also use this function for your testing, but **do not include it in your submitted answer**.
- Hint: Python supports *mutual recursion*, which means you can write two functions that refer to one another.

4 Marking

Marks are available as follows:

Criterion	**Marks**
Q1	15
Q2	15
Q3: compute determinant of 5×5 matrices	10
Q3: invert 2×2 non-singular matrices	10
Q3: invert 5×5 non-singular non-sparse matrices	10
Q3: invert 7×7 non-singular sparse matrices efficiently	20
Q3: handle exception cases	15
Q3: Well-formatted code with appropriate comments	5
TOTAL available	**100**